Ingredients

Hamilton principle

$$\delta\mathscr{A}^\star[u,p] + \forall\delta p, \begin{cases}\forall\delta u = 0 \text{ on } \partial\omega_t\\ \forall\delta u = 0 \text{ on } (\partial\omega_X)^{\text{Dirichlet}}\end{cases}$$

Thermal capacity

$$c = T\frac{\partial S}{\partial T} = \frac{\partial \hat{E}_{\text{tot}}}{\partial T}$$

Helmholtz free energy

$$W(\varepsilon, T, \tilde{\alpha}) = \hat{E}_{\text{int}}(\varepsilon, T, \tilde{\alpha}) - TS(\varepsilon, T, \tilde{\alpha})$$

$\downarrow \mathrm{d}/\mathrm{d}t$

Gibbs' law

$$\frac{\mathrm{d}W}{\mathrm{d}t} = \frac{\mathrm{d}\hat{E}_{\text{int}}}{\mathrm{d}t} - T\frac{\mathrm{d}S}{\mathrm{d}t} - \dot{T}$$

Fourier's law

$$q = -K\frac{\partial T}{\partial X}$$

Biot's law

$$\frac{\partial W}{\partial \tilde{\alpha}} + \frac{\partial \Phi}{\partial \dot{\tilde{\alpha}}} = \tilde{0}$$

State Laws

$$\frac{\partial p}{\partial t} = -\frac{\partial \hat{E}_{\text{tot}}}{\partial u} + \frac{\partial}{\partial X}\left(\frac{\partial \hat{E}_{\text{tot}}}{\partial \varepsilon}\right) - \frac{\partial}{\partial X}\left(\frac{\partial S}{\partial \varepsilon}\left(\frac{\partial S}{\partial T}\right)^{-1}\frac{\partial \hat{E}_{\text{tot}}}{\partial T}\right)$$

$$\frac{\partial \varepsilon}{\partial t} = \frac{\partial}{\partial X}\left(\frac{\partial \hat{E}^\star_{\text{tot}}}{\partial p}\right)$$

$\uparrow \partial/\partial X$

$$\frac{\partial u}{\partial t} = \frac{\partial \hat{E}^\star_{\text{tot}}}{\partial p}$$

$$\begin{aligned}\frac{\partial T}{\partial t} = &-\left(\frac{\partial S}{\partial T}\right)^{-1}\frac{\partial S}{\partial \varepsilon}\frac{\partial}{\partial X}\left(\frac{\partial \hat{E}_{\text{tot}}}{\partial p}\right)\\ &+\frac{1}{(\rho c)^2}K\frac{\partial}{\partial X}\left(T^2\frac{\partial}{\partial X}\left(\frac{\partial S}{\partial T}\right)\right)\\ &-\frac{T}{(\rho c)^2}\left(\frac{\partial \hat{E}_{\text{int}}}{\partial \tilde{\alpha}}\right)^{\mathsf{T}}\tilde{\tilde{\mathsf{V}}}^{-1}\frac{\partial \hat{E}_{\text{int}}}{\partial \tilde{\alpha}}\frac{\partial S}{\partial T}\\ &-\frac{T}{\rho c}\left(\frac{\partial \hat{E}_{\text{int}}}{\partial \tilde{\alpha}}\right)^{\mathsf{T}}\tilde{\tilde{\mathsf{V}}}^{-1}\frac{\partial S}{\partial \tilde{\alpha}}\end{aligned}$$

$$\frac{\partial \tilde{\alpha}}{\partial t} = -\tilde{\tilde{\mathsf{V}}}^{-1}\frac{\partial \hat{E}_{\text{int}}}{\partial \tilde{\alpha}}\frac{T}{\rho c}\frac{\partial S}{\partial T} + T\tilde{\tilde{\mathsf{V}}}^{-1}\frac{\partial S}{\partial \tilde{\alpha}}$$

Evolution

$$\begin{aligned}\frac{\partial F}{\partial t} = &\frac{\partial F}{\partial u}\frac{\partial u}{\partial t} +\\ &+\frac{\partial F}{\partial \varepsilon}\frac{\partial \varepsilon}{\partial t} +\\ &+\frac{\partial F}{\partial p}\frac{\partial p}{\partial t} +\\ &+\frac{\partial F}{\partial T}\frac{\partial T}{\partial t} +\\ &+\left(\frac{\partial F}{\partial \tilde{\alpha}}\right)^{\mathsf{T}}\frac{\partial \tilde{\alpha}}{\partial t}\end{aligned}$$

DGBA Bracket Structure

$$\frac{\mathrm{d}\mathscr{F}}{\mathrm{d}t} = \{\mathscr{F}, \hat{\mathscr{E}}_{\text{tot}}\} + (\mathscr{F}, \mathscr{S}) + \text{boundary terms}$$

$$\{\mathscr{F}, \hat{\mathscr{E}}_{\text{tot}}\} = \int_{\omega_X}\begin{pmatrix}\frac{\delta F}{\delta u}\\ \frac{\partial F}{\partial p}\\ \frac{\partial F}{\partial T}\\ \frac{\partial F}{\partial \tilde{\alpha}}\end{pmatrix}\begin{pmatrix}0 & 1 & 0 & 0\cdots 0\\ -1 & 0 & \left(\frac{\partial S}{\partial T}\right)^{-1}\frac{\partial S}{\partial \varepsilon}\frac{\partial \square_{\text{left}}}{\partial X} & \vdots \quad \vdots\\ 0 & -\left(\frac{\partial S}{\partial T}\right)^{-1}\frac{\partial S}{\partial \varepsilon}\frac{\partial \square_{\text{right}}}{\partial X} & 0 & \vdots \quad \vdots\\ 0\cdots & \cdots & \cdots & 0\cdots 0\\ \vdots & & & \vdots \quad \vdots\\ 0\cdots & \cdots & \cdots & 0\cdots 0\end{pmatrix}\begin{pmatrix}\frac{\delta \hat{E}_{\text{tot}}}{\delta u}\\ \frac{\partial \hat{E}_{\text{tot}}}{\partial p}\\ \frac{\partial \hat{E}_{\text{tot}}}{\partial T}\\ \frac{\partial F}{\partial \tilde{\alpha}}\end{pmatrix}\mathrm{d}\omega_X$$

$$(\mathscr{F}, \mathscr{S}) = \int_{\omega_X}\begin{pmatrix}\frac{\delta F}{\delta u}\\ \frac{\partial F}{\partial p}\\ \frac{\partial F}{\partial T}\\ \frac{\partial F}{\partial \tilde{\alpha}}\end{pmatrix}\begin{pmatrix}0 & 0 & 0 & 0 & \tilde{0}^{\mathsf{T}}\\ 0 & 0 & 0 & 0 & \tilde{0}^{\mathsf{T}}\\ 0 & 0 & 0 & 0 & \tilde{0}^{\mathsf{T}}\\ 0 & 0 & 0 & \frac{\partial \square_{\text{left}}}{\partial X}K\left(\frac{T}{\rho c}\right)^2\frac{\partial \square_{\text{right}}}{\partial X} + \frac{T}{(\rho c)^2}\left(\frac{\partial \hat{E}_{\text{int}}}{\partial \tilde{\alpha}}\right)^{\mathsf{T}}\tilde{\tilde{\mathsf{V}}}^{-1}\frac{\partial \hat{E}_{\text{int}}}{\partial \tilde{\alpha}} & -\frac{T}{\rho c}\left(\frac{\partial \hat{E}_{\text{int}}}{\partial \tilde{\alpha}}\right)^{\mathsf{T}}\tilde{\tilde{\mathsf{V}}}^{-1}\\ \tilde{0} & \tilde{0} & \tilde{0} & -\frac{T}{\rho c}\tilde{\tilde{\mathsf{V}}}^{-1}\frac{\partial \hat{E}_{\text{int}}}{\partial \tilde{\alpha}} & T\tilde{\tilde{\mathsf{V}}}^{-1}\end{pmatrix}\begin{pmatrix}\frac{\delta S}{\delta u}\\ \frac{\partial S}{\partial p}\\ \frac{\partial S}{\partial T}\\ \frac{\partial S}{\partial \tilde{\alpha}}\end{pmatrix}\mathrm{d}\omega_X$$

$$\begin{aligned}\text{boundary terms} = &-\int_{\partial\omega_X}\frac{\partial F}{\partial T}K\left(\frac{T}{\rho c}\right)^2\frac{\partial}{\partial X}\left(\frac{\partial S}{\partial T}\right)\mathrm{d}(\partial\omega_X) + \int_{\partial\omega_X}\frac{\partial F}{\partial p}\left(\frac{\partial \hat{E}_{\text{tot}}}{\partial \varepsilon} - \left(\frac{\partial S}{\partial T}\right)^{-1}\frac{\partial S}{\partial \varepsilon}\frac{\partial \hat{E}_{\text{tot}}}{\partial T}\right)\mathrm{d}(\partial\omega_X) +\\ &+\int_{\partial\omega_X}\frac{\partial F}{\partial \varepsilon}\left(\frac{\partial u}{\partial t} - \frac{\partial \hat{E}_{\text{tot}}}{\partial p}\right)\mathrm{d}(\partial\omega_X)\end{aligned}$$